

SPADMIC block-owned CSR banks

ABI 1.0 register ownership, write admission, W1C fault handling, and saturating diagnostics.

TOP/rtl/spadmic_csr_banks.sv + TOP/rtl/spadmic_csr_map_pkg.sv

System bank

ID and ABI version
atomic GLOBAL_CTRL
safe-idle and access diagnostics
maintenance counter clear

TDC R/Y/B banks

per-axis status
W1C FIFO fault
saturating error count

Position bank

cluster/raw mode
gap and minimum span
drop fault/count

Event bank

event lifecycle and masks
snapshot/reset configuration
R/Y/B snapshot readback

Matrix bank

column command and 64-bit data
readback and validity
W1C command fault

TX bank

bundle/FIFO/DDR status
fixed FIFO geometry
missing/overflow faults and counts

PLL bank

PLL control image
lock status
disabled-and-idle write gate

Analog bank

SLVS/RX controls
explicit active polarity
disabled-and-idle write gate

Shared bank policy

Configuration writes require global disabled and safe idle.
GLOBAL_CTRL validates and commits a complete image atomically.
Faults are W1C; maintenance clears counters only while disabled and idle.